Artificial Intelligence (AI):

Unit-1 Introduction

1.1 Definitions

Intelligence: It is the capacity to learn and solve problems. It is to solve novel problems, act rationally and act like humans.

– the ability to reason

– the ability to understand

– the ability to create

– the ability to learn from experience

– the ability to plan and execute complex tasks

Artificial Intelligence (AI): The branch of computer science concerned with creating machines capable of performing tasks that typically require human intelligence. This includes reasoning, learning, problem-solving, perception and language understanding.

["Giving machines ability to perform tasks normally associated with human intelligence."]

AI definitions can be categorized into four dimension, they are as follows:

- **Systems that think like humans**
- **Systems that think rationally**
- **Systems that act like humans**
- **System that act rationally**

Acting Humanly: Turning Test Approach

A Turing Test is a method of inquiry in artificial intelligence (AI) for determining whether or not a computer is capable of thinking like a human being. The test is named after Alan Turing, the founder of the Turning Test and an English computer scientist, cryptanalyst, mathematician and theoretical biologist. Turing proposed that a computer can be said to possess artificial intelligence if it can mimic human responses under specific conditions.

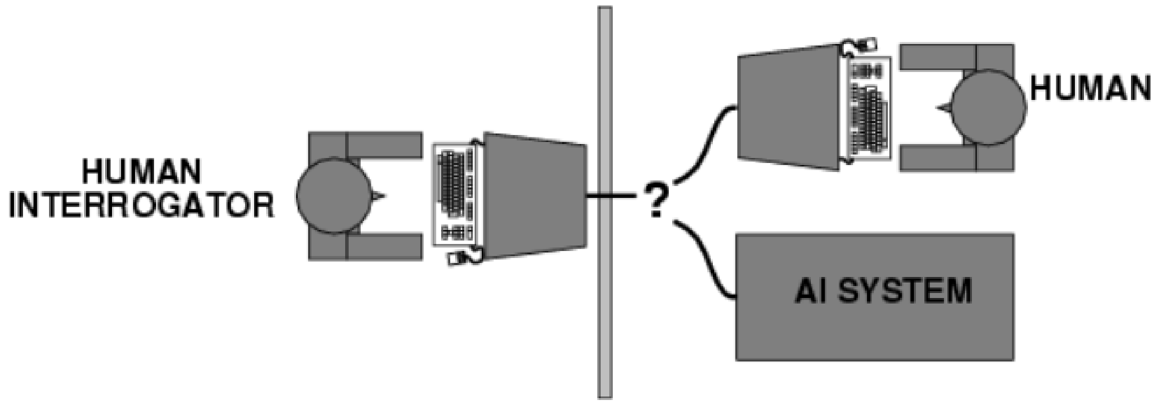


Fig: The above figure Depicts Turning Test

Thinking Humanly: The Cognitive Modeling Approach

When program thinks like a human, it must have some way of determining how humans think. Two ways to get to know the actual workings of human minds namely:

- Through Introspection (Trying to catch one's own thoughts)
- Through Psychological Experiments.

Thinking Rationally

The Laws of Thought Approach Right thinking that is, irrefutable (imposible to denied) reasoning processes.

Acting Rationally

The Rational Agent Approach An agent is something that perceives and acts. AI is looked upon as a study that deals with the construction and study of the rational agents. Forming the right inferences is sometimes part of being a rational agent. One way to act rationally is to reason logically the conclusion that might achieve the goal and then only to act on the conclusion.

1.2 Brief History of AI, Turing Test

History of AI:

1950s: Alan Turing proposes the Turing Test in his paper "Computing Machinery and Intelligence", questioning "Can machines think?"

1956: The term "Artificial Intelligence" is coined at the Dartmouth Conference, marking the birth of AI as a field.

1960s-70s: Development of early AI programs (e.g., ELIZA, an early natural language processor).

1980s: Introduction of expert systems (software that mimics human decision-making).

1997: IBM's Deep Blue defeats chess champion Garry Kasparov.

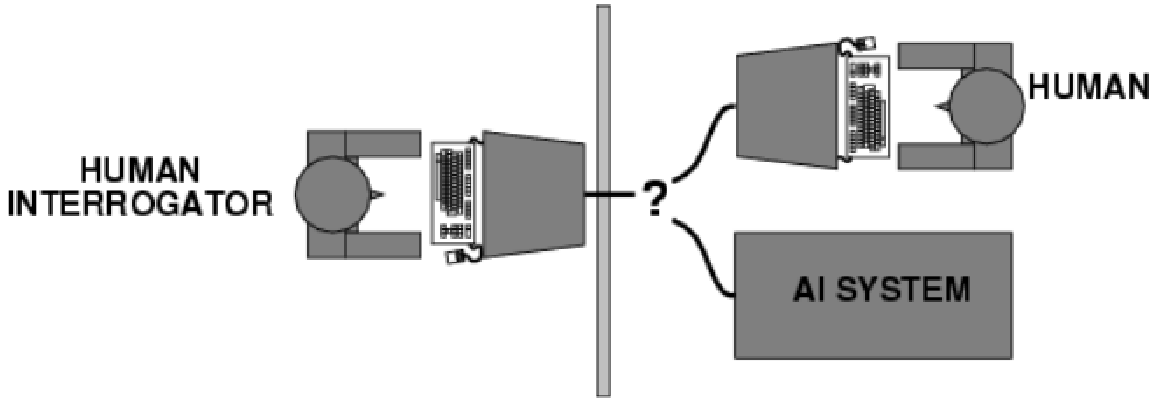
2010s-Present: Emergence of machine learning and deep learning technologies; notable achievements like AlphaGo defeating Go champion Lee Sedol.

Turing Test:

A test to evaluate a machine's ability to exhibit intelligent behavior indistinguishable from that of a human. If a human interrogator cannot reliably distinguish between the machine and a human, the machine passes the test.

To pass a Turing test, a computer must have following capabilities:

- ***Natural Language Processing:*** Must be able to communicate successfully in English
- ***Knowledge representation:*** To store what it knows and hears.
- ***Automated reasoning:*** Answer the Questions based on the stored information.
- ***Machine learning:*** Must be able to adapt in new circumstances.



The Turing test, proposed by Alan Turing (1950) was designed to convince the people that whether a particular machine can think or not. He suggested a test based on indistinguishability from undeniably intelligent entities- human beings. The test involves an interrogator who interacts with one human and one machine. Within a given time the interrogator has to find out which of the two the human is, and which one the machine.

1.3 AI and Related Fields

- ***Machine Learning (ML):*** A subset of AI focused on algorithms that improve automatically through experience.
- ***Natural Language Processing (NLP):*** Enables machines to understand and interact using human language.
- ***Robotics:*** The design and creation of robots that can perform tasks autonomously.
- ***Computer Vision:*** Enabling machines to interpret and make decisions based on visual data.
- ***Cognitive Science:*** The interdisciplinary study of mind and intelligence, overlapping with AI in understanding learning and problem-solving.

1.4 Goals and Challenges of AI

Goals:

- Create systems that can reason, learn and adapt.
- Achieve general AI capable of performing any intellectual task a human can do.
- Enhance automation, efficiency, and decision-making across industries.

Challenges:

- ***Ethical concerns:*** Bias in algorithms, data privacy and job displacement.
- ***Complexity:*** Modeling human intelligence is intricate and difficult.
- ***Resource Intensity:*** High computational power and data requirements.
- ***Explainability:*** AI decisions can be difficult to interpret (especially in deep learning).

1.5 Applications of AI

Artificial Intelligence has various applications in today's society. It is becoming essential for today's time because it can solve complex problems with an efficient way in multiple industries, such as Healthcare, entertainment, finance, education, etc. AI is making our daily life more comfortable and fast. Following are some sectors which have the application of Artificial Intelligence:

AI in Astronomy:

- Artificial Intelligence can be very useful to solve complex universe problems.
- AI technology can be helpful for understanding the universe such as how it works, origin, etc.

AI in Healthcare: (Diagnostics, personalized medicine, robotic surgeries***)***

- In the last, five to ten years, AI becoming more advantageous for the healthcare industry and going to have a significant impact on this industry.
- Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization.

AI in Gaming

- AI can be used for gaming purpose. The AI machines can play strategic games like chess, where the machine needs to think of a large number of possible places.

AI in Finance (**Fraud detection, algorithmic trading, risk management**)

- AI and finance industries are the best matches for each other.
- The finance industry is implementing automation, chat-bot, adaptive intelligence, algorithm trading and machine learning into financial processes.

AI in Data Security

- The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world. AI can be used to make your data more safe and secure. Some examples such as AEG bot, AI2 Platform, are used to determine software bug and cyber-attacks in a better way.

AI in Social Media

- Social Media sites such as Facebook, Twitter, and Snap chat contain billions of user profiles, which need to be stored and managed in a very efficient way. AI can organize and manage massive amounts of data. AI can analyze lots of data to identify the latest trends, hash tag, and requirement of different users.

AI in Travel &Transport

- AI is becoming highly demanding for travel industries. AI is capable of doing various travel related works such as from making travel arrangement to suggesting the hotels, flights and best routes to the customers.
- Travel industries are using AI-powered chat-bots which can make human-like interaction with customers for better and fast response.

AI in Automotive Industry: (**Self-driving cars, predictive maintenance**)

- Some Automotive industries are using AI to provide virtual assistant to their user for better performance. Such as Tesla has introduced Tesla-Bot, an intelligent virtual assistant.
- Various Industries are currently working for developing self-driven cars which can make your journey more safe and secure.

AI in Robotics

- Artificial Intelligence has a remarkable role in Robotics. Usually, general robots are programmed such that they can perform some repetitive task but with the help of AI, we can create intelligent robots which can perform tasks with their own experiences without pre-programmed.
- Humanoid Robots are best examples for AI in robotics, recently the intelligent Humanoid robot named as Erica and Sophia has been developed which can talk and behave like humans.

AI in Entertainment (**Recommendation systems (e.g., Netflix, Spotify)**)

- We are currently using some AI based applications in our daily life with some entertainment services such as Netflix or Amazon. With the help of ML/AI algorithms, these services show the recommendations for programs or shows.

AI in Agriculture (**Precision farming, crop monitoring**)

- Agriculture is an area which requires various resources, labor, money and time for best result. Now a day's agriculture is becoming digital and AI is emerging in this field.
- Agriculture is applying AI as agriculture robotics, solid and crop monitoring, predictive analysis. AI in agriculture can be very helpful for farmers.

AI in E-commerce

- AI is providing a competitive edge to the e-commerce industry, and it is becoming more demanding in the e-commerce business. AI is helping shoppers to discover associated products with recommended size, color or even brand.

AI in Education (Personalized learning platforms, AI Tutors***)***

- AI can automate grading so that the tutor can have more time to teach. AI chat-bot can communicate with students as a teaching assistant.
- AI in the future can be work as a personal virtual tutor for students which will be accessible easily at any time and any place.